THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 70°	angle sum is 180°
2	**(b)** 78°	angle sum is 180°
3	**(a)** 60°	exterior angle = sum of the two interior opposite angles
4	**(d)** Yes	sum of any two sides must exceed the third
5	**(b)** No	sum of any two sides must exceed the third
6	**(b)** 55°	base angles are equal
7	**(c)** right-angled	definition
8	**(b)** orthocentre	concurrent
9	**(b)** hypotenuse	opposite the right angle
10	**(c)** greater than each interior opposite angle	sum of the two opposite angles

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The three interior angles of a triangle add to 180° (2).	2
12	60° (2).	2
13	Altitude is perpendicular from a vertex to the opposite side (1); median joins vertex to midpoint of opposite side (1).	1+1
14	No; 4 + 5 < 10 (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	x = 20 (1); 40°, 60°, 80° (2).	1+2
16	65° (1.5); 70° (1.5).	1½+1½
17	9x = 180; x = 20 (1.5); 20°, 80°, 80° (1.5).	1½+1½
	OR: 5 + 6 < 12 so no (1.5); shorten 12 below 11 (1.5).	1½+1½
18	Draw 5 cm base (1); arcs of 5 cm from both ends (1); join the point (1).	1+1+1
	OR: 6x = 180, x = 30 (1.5); 40°, 60°, 80° (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) The three corners form a straight line (2). (ii) 70° (2). (iii) acute scalene (1).	2+2+1
	OR: (i) 40° each (1). (ii) 80° (1). (iii) Obtuse isosceles (1). (iv) 14 m (2).	1+1+1+2
20	(i) Base 6; arcs 4 and 5 (2). (ii) Set square (2). (iii) Intersection of altitudes (1).	2+2+1
	OR: (i) Sum of any two sides exceeds the third (1). (ii) No, yes, yes (3). (iii) 3 + 4 < 8 (1).	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 70°	angle sum is 180°
2	**(b)** 65°	angle sum is 180°
3	**(c)** 70°	exterior angle = sum of the two interior opposite angles
4	**(a)** No	sum of any two sides must exceed the third
5	**(b)** Yes	sum of any two sides must exceed the third
6	**(d)** 50°	base angles are equal
7	**(a)** obtuse-angled	definition
8	**(c)** centroid	balance point
9	**(a)** one	sum is 180°
10	**(b)** 21 cm	3 × 7

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	60°; angles equal and sum 180° (2).	2
12	An exterior angle equals the sum of the two interior opposite angles (2).	2
13	Equilateral, isosceles, scalene (2).	2
14	Acute-angled, right-angled, obtuse-angled (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Place a set square on a side (1); slide so it touches the opposite vertex (1); draw perpendicular (1).	1+1+1
16	3x + 30 = 180; x = 50 (1.5); largest 70° (1.5).	1½+1½
17	Each vertex joins the midpoint of the opposite side (1.5); gives three medians (1.5).	1½+1½
	OR: x + x + 20 = 90 (1.5); 35° and 55° (1.5).	1½+1½
18	(36 − 10) ÷ 2 = 13 cm (3).	3
	OR: 9 + 16 = 25 (2); yes (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 70° (1). (ii) Opposite the largest angle: AB (1). (iii) 110° (1). (iv) acute-angled scalene (2).	1+1+1+2
	OR: (i) 30°, 60°, 90° (2). (ii) right-angled (1). (iii) 90° (2).	2+1+2
20	(i) Exterior + interior adjacent = 180 (2). (ii) 90° (1.5). (iii) 90° (1.5).	2+1½+1½
	OR: (i) Steps (2). (ii) 4 cm (1.5). (iii) 6 cm² (1.5).	2+1½+1½

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.